

Assignment

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Course Code:- ITA0612

Course Name:- machine learning.

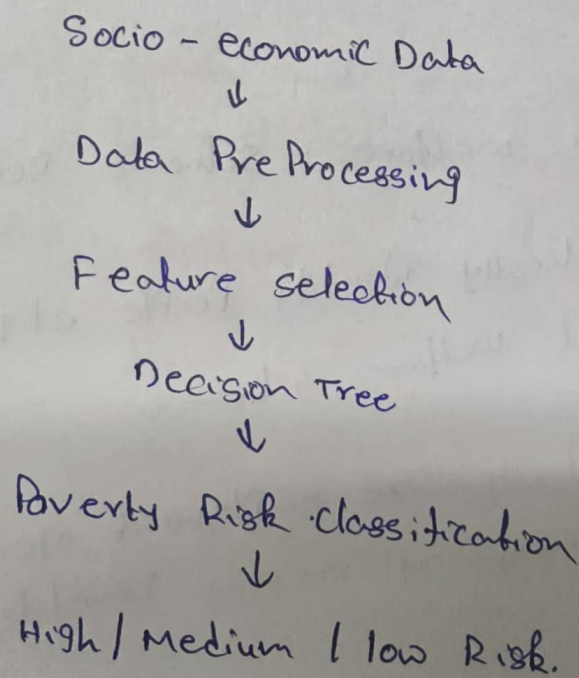
1) Machine Learning to Identity Poverty - Prone Regions.

Problem:- Identity regions where People are at high risk of Poverty

Input Data:- Income, employment, education, housing, Population, healthcare access and living Conditions.

Algorithm:- Decision Tree.

working :-



The Decision Tree divides regions using important socio economic factors. Regions with low income, unemployment and poor living conditions can be classified as Poverty - Prone.

Outcome:- Government can identify high risk regions and Prioritize welfare Programs.

Advantages:- Easy to understand interpretable and useful for Policy Planning.

Limitations:- Poor-quality or incomplete Socio-economic data can reduce accuracy.

Conclusion:- ML can support targeted Poverty-reduction Programs.

2) Identifying Eligible welfare - scheme Beneficiaries;

Problem:- Automatically identify People eligible for Government welfare schemes.

Features:- Income, family size, employment, age, disability status, housing and existing benefits.

The model learns from Previously verified applications and Predicts eligibility for new applications.

Algorithm:- Decision Tree.

Applicant Data



Data cleaning



Feature selection



Decision tree model



Eligibility Prediction



Eligible/not Eligible.

Outcomes:- Faster and more consistent identification of beneficiaries.

Advantages:- Reduces manual work and Processing time.

Limitations:- Bias in historical application data may produce unfair predictions.

Conclusion:- ML can improve the efficient distribution of welfare benefits.

3) Crop Yield Prediction:-

Problem:- Predict crop yield using historical agricultural data.

Features:- Rainfall, temperature, soil type, fertilizer, irrigation, Crop type and Previous yield.

Algorithm:- Decision Tree Regression.

Historical, Agricultural Data



Data PreProcessing



Feature Selection



Decision Tree Regression



Yield Prediction



Estimated Crop Yield.

Outcomes:- Farmers can plan irrigation, fertilizers and harvesting.

Advantages:- Supports agricultural planning and food security decisions.

Limitations:- Unexpected weather conditions can affect predictions.

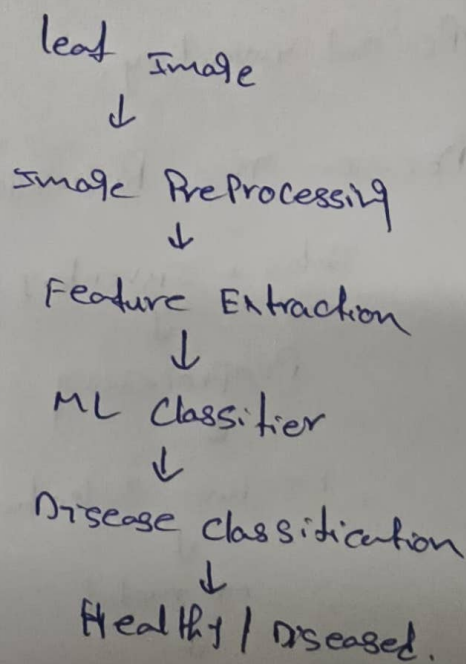
Conclusion:- ML-based Yield Prediction can
Contribute to food security.

4) Crop Disease Detection Using Leaf Images:-

Problem:- Detect diseases from crop leaf images.

Input:- Images of healthy and diseased leaves.

Algorithm:- Decision Tree can be used with extracted
image features, image-based systems can also
use classification techniques.



Outcomes:- Early disease identification.

Advantages:- Helps farmers take timely action.

Limitations:- Poor lighting and image quality may affect prediction.

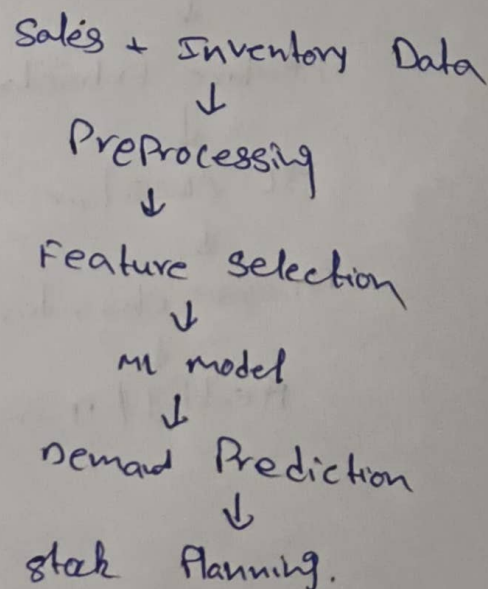
Conclusion:- ML can help reduce crop losses through early disease detection.

5) Reducing Food wastage in Supermarkets

Problem:- Predict Product demand to Prevent overstocking and wastage.

Features:- Sales history, Season, Price, holidays, Product type and inventory.

Algorithm:- Decision Tree Regression.



Outcome:- stores maintain appropriate inventory levels.

Advantages:- Reduces food waste and improves inventory management.

Limitation:- sudden changes in customer demand may reduce accuracy.

Conclusion:- ML can help supermarkets reduce unnecessary food wastage.